A Multi-Country Spatiotemporal Machine Learning Framework for Methane Hotspot Detection Using Satellite Observations

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***Abstract*—Methane (CH_4) is a potent greenhouse gas with major implications for climate change and atmospheric chemistry. Detecting methane emission hotspots remains challenging due to spatial heterogeneity, atmospheric transport processes, and limitations in observational data. This study presents a scalable, multi-country framework for methane hotspot detection based on satellite observations and machine learning. A spatiotemporal dataset was constructed using Google Earth Engine by integrating TROPOMI methane concentrations, Sentinel-2 surface reflectance, ERA5 wind fields and additional geospatial variables on a hexagonal grid across Europe from 2021 to 2025. A LightGBM-based classification model was trained to identify methane hotspots using both methane-derived and environmental features. To ensure robustness and avoid data leakage, a global normalization strategy was applied across all regions. An ablation study shows that methane-derived features are the primary drivers of model performance, while environmental variables provide complementary explanatory information. Cross-country validation demonstrates strong generalization, with consistently high ROC-AUC scores and more variable PR-AUC values reflecting regional emission characteristics. Overall, the proposed framework enables scalable methane monitoring and offers insights into the interaction between atmospheric conditions and surface drivers of emissions. This work contributes a robust pipeline for satellite-based environmental monitoring and highlights the potential for future integration with hyperspectral data, such as EMIT, to further improve detection capabilities.**

***Keywords*—methane detection, machine learning, satellite observations**

I. Introduction

Methane (CH_4) is a major contributor to anthropogenic climate change, with a global warming potential significantly higher than carbon dioxide over short time horizons [1]. Due to its relatively short atmospheric lifetime, methane mitigation offers substantial potential for near-term climate benefits. Consequently, the detection and monitoring of methane emission hotspots is critical for climate policy, regulatory enforcement and emission inventory verification. Satellite remote sensing has become an essential tool for large-scale monitoring of atmospheric methane. However, accurate identification of localized emission hotspots remains challenging due to atmospheric transport processes, spatial heterogeneity and limitations in observational data. In addition, integrating multi-source geospatial datasets and ensuring robust model generalization across regions introduces further methodological complexity. Recent advances in machine learning have improved the detection of environmental anomalies, but important challenges remain, particularly in ensuring model generalization across geographic regions, preventing data leakage, and integrating multi-source datasets with differing spatial and temporal resolutions. This study introduces a unified spatiotemporal framework for methane hotspot detection across multiple European countries, combining a scalable feature engineering pipeline built on cloud-based geospatial processing with a leakage-free global normalization strategy. The framework further incorporates a machine learning approach with spatially aware validation, supported by an ablation study to evaluate the contribution of methane-derived features and a cross-country generalization analysis to assess robustness across regions.

II. State-of-The-Art

A. *Satellite-based methane monitoring*

Satellite-based methane monitoring has advanced significantly in recent years, enabling improved detection and quantification of emission sources across multiple spatial scales. Sentinel-2 has been widely studied for its potential in methane detection, particularly for industrial point sources. Zhang et al. [2] investigated methane emission detection and quantification from oil and gas production using Sentinel-2 data, demonstrating its applicability under certain observational conditions. Similarly, Gorroño et al. [3] analyzed the potential of Sentinel-2 for monitoring methane point emissions, highlighting both opportunities and limitations of multispectral detection approaches. Irakulis-Loitxate et al. [4] further demonstrated methane point-source detection using high-resolution satellite imagery, emphasizing the importance of spatial resolution for reliable detection. Recent work by Schuit et al. [5] introduced an automated framework for detecting and monitoring methane super-emitters using TROPOMI data in combination with high-resolution satellite observations. Their approach enables large-scale identification of emission plumes and persistent hotspot regions, demonstrating the feasibility of global methane monitoring systems. In addition, comprehensive review studies have summarized recent progress in satellite-based methane monitoring and retrieval methods [6], [7], highlighting ongoing challenges related to retrieval accuracy, atmospheric interference and scalability.

B. *Machine learning for methane detection*

Machine learning and artificial intelligence techniques have increasingly been applied to methane detection and atmospheric monitoring tasks, particularly in the context of multispectral satellite imagery. Recent work by Rouet-Leduc et al. [8] introduced a vision transformer-based approach for automatic detection of methane emissions in multispectral satellite imagery, demonstrating the potential of deep learning for automated plume identification. Růžička et al. [9] developed operational machine learning frameworks for remote spectroscopic detection of methane point sources, enabling scalable processing of satellite observations. In addition, the MARS-S2L framework proposed by Allen et al. [10] demonstrates the application of artificial intelligence for methane detection using Sentinel-2 imagery. The approach focuses on automated detection pipelines for methane plume identification and highlights the potential of AI-driven methods for scalable satellite-based monitoring. However, such frameworks are primarily designed for detection and retrieval tasks based on individual satellite scenes or localized observations. They generally focus on identifying methane plumes or emission signatures in a scene-wise manner, without explicitly addressing broader challenges such as cross-regional generalization, systematic data leakage prevention, or the fusion of heterogeneous multi-source datasets with differing spatial and temporal resolutions.

C. *Research Gap*

Although satellite-based methane detection and machine learning approaches have advanced significantly, key gaps remain. Existing methods are often developed for limited geographic regions, restricting their generalizability. Recent AI-based frameworks such as MARS-S2L [10] and plume detection approaches (e.g., Schuit et al. [5]) demonstrate effective event detection but primarily focus on scene-based retrieval rather than generalized hotspot modeling across regions. Moreover, challenges related to data leakage, spatial dependencies in evaluation, and the integration of heterogeneous multi-source datasets remain insufficiently addressed. These limitations motivate the need for a unified framework for robust and scalable methane hotspot detection across diverse geographic domains.

III. Datasets and Methodology

A. *Spatiotemporal Feature Engineering*

A spatiotemporal methane feature dataset was constructed using Google Earth Engine by integrating multi-source satellite and reanalysis data on a hexagonal grid. Monthly composites were generated for the period 2021–2025, with all observations aggregated at the grid-cell level to ensure spatial consistency across datasets. The dataset incorporates four main data sources: TROPOMI methane column concentrations, Sentinel-2 surface reflectance, ERA5 reanalysis wind fields and auxiliary geospatial layers including topographic and infrastructure information. All variables were spatially and temporally aligned prior to aggregation. Methane-related features were derived from TROPOMI observations and include the mean concentration, standard deviation, standardized anomalies (z-scores) and a persistence metric defined as the frequency of exceedance above a 2σ threshold. These features capture both baseline emission levels and temporal variability of methane signals. In addition, a set of optical indices was computed from cloud-masked Sentinel-2 imagery, including NDVI, NDWI, NDMI, NDBI and BSI, to characterize surface properties potentially related to emission sources and land use patterns. Meteorological context was provided by ERA5-derived wind speed and wind direction, enabling the characterization of atmospheric transport conditions. All features were aggregated to the hexagonal grid using mean-based reducers, ensuring consistent spatial summarization across heterogeneous input resolutions.

B. *Global Normalization*

To ensure consistency across regions and to mitigate the risk of data leakage, a global normalization strategy was applied. Specifically, feature-wise statistics were computed over the entire dataset, spanning all countries and the full temporal range 2021–2025. The global mean μ and standard deviation σ were then used to normalize each feature as follows:

$$X_{\text{norm}} = \frac{X - \mu}{\sigma}$$

This approach ensures that all input features are placed on a consistent scale, independent of regional or temporal variations in their distributions. By using globally computed normalization parameters, the method avoids region-specific scaling effects that could otherwise introduce implicit information leakage between training and test domains. Consequently, this normalization strategy improves comparability across heterogeneous geographic regions and supports robust cross-country generalization of the machine learning model (see Fig. 1).

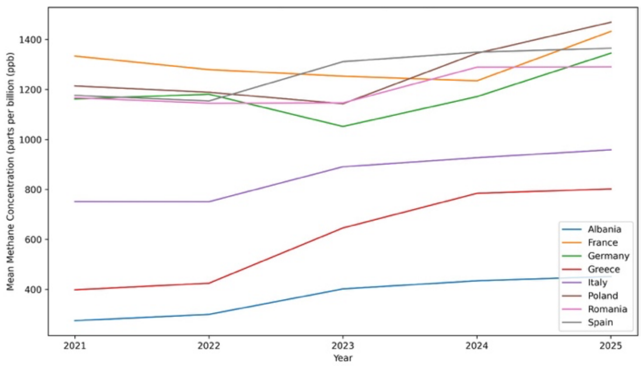

Fig. 1. Yearly Methane Trends by Country

C. *Model Development*

A LightGBM-based classification model was developed to predict methane emission hotspots using the engineered spatiotemporal features. LightGBM was selected due to its efficiency in handling large-scale datasets and its strong performance on structured tabular data. To ensure a robust evaluation and prevent spatial overfitting, a spatially aware data splitting strategy was employed. Specifically, a GroupShuffleSplit approach was used, where samples were grouped by their corresponding hexagonal grid identifiers. This ensures that all observations from a given spatial unit are assigned exclusively to either the training or testing set. By enforcing spatial separation between training and testing data, this strategy mitigates spatial data leakage and prevents the model from exploiting location-specific patterns that do not generalize beyond the training regions. As a result, the evaluation more accurately reflects the model's ability to generalize to unseen geographic areas.

D. *Ablation Study*

An ablation study was conducted to assess the contribution of methane-derived features to the overall model performance. Two model configurations were evaluated under identical training and validation settings:

- **Model A (Full Model):** Includes both methane-derived features and auxiliary environmental variables
- **Model B (Strict Model):** Excludes all methane-derived features and relies solely on environmental predictors

This experimental design enables a systematic comparison between direct methane observations and indirect environmental signals. By isolating the contribution of methane-specific features, the study quantifies their impact on predictive performance and evaluates the extent to which environmental variables alone can capture methane emission patterns. All other components of the pipeline, including data preprocessing, normalization and spatial validation strategy, were kept constant to ensure a controlled and fair comparison between the two configurations.

IV. RESULTS

A. Model Performance

The full model achieved strong predictive performance across all evaluation metrics, with a ROC-AUC of approximately 0.99, a PR-AUC of approximately 0.97 and an F1 score of approximately 0.98. These results indicate a high capability to accurately detect methane emission hotspots. In contrast, the constrained model, excluding methane-derived features, showed substantially lower performance, with a ROC-AUC of approximately 0.81, a PR-AUC of approximately 0.02 and an F1 score of approximately 0.04. While the ROC-AUC remains moderately high, the very low PR-AUC and F1 score indicate poor performance in identifying the positive class, particularly under class imbalance conditions. This discrepancy highlights the critical role of methane-derived features for reliable hotspot detection. The results suggest that environmental variables alone are insufficient to capture methane emission patterns with high precision and recall and primarily provide complementary rather than primary predictive signals.

B. Cross-Country Generalization and Ablation Analysis

Model performance was evaluated across eight European countries to assess generalization under varying conditions. ROC-AUC remained consistently high across regions, indicating stable discrimination, while PR-AUC varied significantly due to differences in emission density and class imbalance. Regions with concentrated industrial emissions showed higher precision–recall performance than those with sparse or diffuse sources, highlighting the model's sensitivity to hotspot distribution. The ablation analysis (see Fig. 2) underscores the importance of methane-derived features.

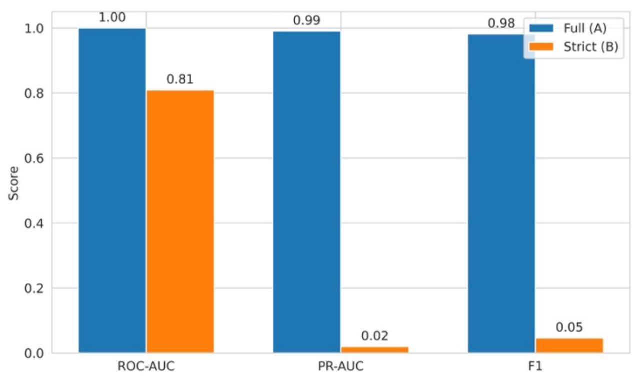

Fig. 2. Ablation Study: Impact of Methane Features

While the full model achieved near-optimal performance, the constrained model - excluding methane observations - showed a substantial decline, particularly in PR-AUC. This indicates that environmental variables alone are insufficient for reliable hotspot detection and primarily serve as complementary contextual signals, whereas methane-derived features are the main drivers of predictive performance.

C. Feature Importance and Spatial Patterns

SHAP analysis (see Fig. 3) reveals that methane-derived features dominate model predictions, with CH_4 mean concentration exhibiting the highest contribution to model output. Interestingly, the standardized anomaly feature (CH_4 z-score) shows minimal variance in SHAP values, suggesting redundancy given the inclusion of raw methane concentration. Environmental predictors, including vegetation indices, wind fields and topographic variables exhibit near-zero marginal contribution, indicating limited independent predictive power. This distribution confirms that the model primarily learns a direct mapping from methane observations to anomaly labels rather than inferring emissions from environmental proxies. The dominance of methane-derived features suggests that the current model primarily performs anomaly reconstruction rather than fully independent emission prediction. While this enables high detection accuracy, it limits the model's ability to generalize in the absence of direct methane observations. Future work will focus on improving physically grounded prediction using higher-resolution environmental and atmospheric transport features.

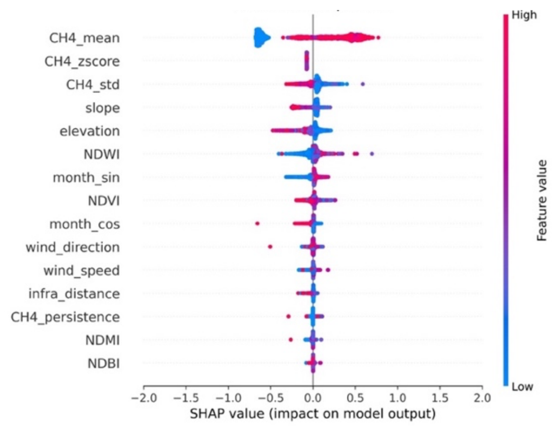

Fig. 3. SHAP Feature Importance

Spatial visualization of predicted methane hotspot probabilities revealed geographically coherent high-intensity clusters across Europe, particularly in industrial and energy-related regions. The hexagonal aggregation ensured consistent spatial representation, while probability thresholding highlighted persistent emissions. To further analyze spatial structure, a density-based clustering approach using DBSCAN was applied (see Fig. 4). High-probability grid cells were grouped based on spatial proximity, identifying coherent emission zones while filtering isolated detections. The results show that hotspots form spatially contiguous clusters corresponding to realistic emission regions. This confirms strong spatial organization of methane emissions and demonstrates the model's ability to capture structured patterns, distinguishing persistent sources from noise or transient signals.

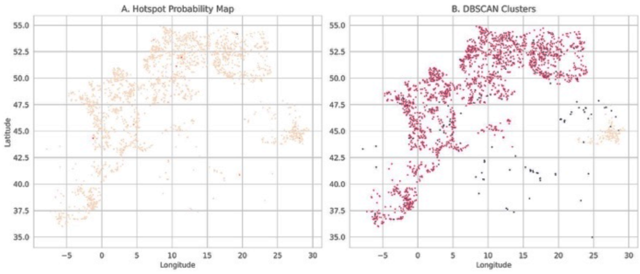

Fig. 4. Visualization of DBSCAN and Hotspot Probability

D. Temporal Dynamics of Methane Hotspots

To assess the temporal behavior of methane emissions, the mean predicted hotspot probability was analyzed over the study period 2021–2025 (see Fig. 5), aggregated across all spatial grid cells. The results reveal a temporally variable but overall stable pattern of methane hotspot activity. While no strong long-term trend is observed, the time series exhibits recurring fluctuations and intermittent peaks, indicating dynamic changes in emission intensity over time. These variations may reflect a combination of seasonal influences, operational variability in emission sources, and atmospheric conditions affecting methane transport and detection. The presence of repeated peaks suggests that methane emissions are not uniformly distributed over time but exhibit structured temporal variability. Although the aggregated nature of this analysis does not resolve regional dynamics, it provides a robust high-level indicator of temporal consistency and variability. The stability of the overall signal, combined with identifiable fluctuations, further supports the reliability and sensitivity of the proposed framework.

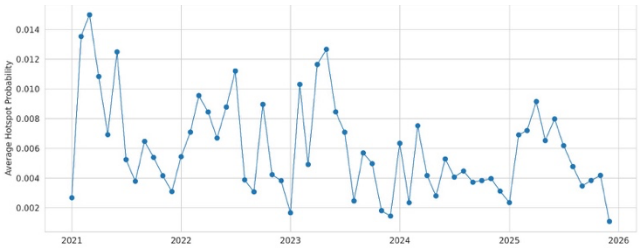

Fig. 5. Temporal Evolution of Methane Hotspots

V. Limitations and Discussion

The results demonstrate that methane hotspot detection is primarily driven by direct atmospheric observations. Methane-derived features, particularly anomaly-based indicators, dominate predictive performance, while environmental variables provide complementary contextual information. These findings are consistent with recent AI-based methane detection frameworks such as MARS-S2L [10], which also emphasize the importance of direct satellite-derived methane signals for reliable detection of emission sources. However, in contrast to scene-based detection frameworks, the proposed approach formulates methane hotspot identification as a spatiotemporal prediction problem across multiple geographic regions. This enables the integration of heterogeneous data sources and supports generalization beyond localized detection settings. Consequently, while detection-oriented frameworks such as MARS-S2L focus on identifying methane plumes in individual observations, the present study addresses the broader problem of robust cross-regional methane hotspot modeling. The proposed global normalization strategy contributes to the robustness of the approach by enabling consistent feature scaling across heterogeneous geographic domains. This facilitates cross-country generalization and reduces the risk of data leakage. However, despite the strong overall performance, several limitations remain. First, the spatial resolution of TROPOMI data constrains the ability to detect small-scale or weak emission sources, particularly in regions with complex emission patterns. Second, cloud contamination in optical satellite imagery introduces data gaps and uncertainty, affecting both feature quality and temporal consistency. Third, the use of monthly temporal aggregation may smooth short-term emission events, limiting the detection of transient or episodic releases. In addition, the current feature space does not fully capture all relevant environmental drivers. Important factors such as temperature, solar angle and atmospheric stability are not explicitly included, although they can significantly influence methane dispersion and retrieval sensitivity. More generally, disentangling emission signals from environmental and atmospheric effects remains a fundamental challenge, particularly when relying on indirect proxies. Another limitation arises from data availability and consistency. The framework relies primarily on open-access datasets, which, while scalable, may be limited in resolution, coverage, and quality compared to proprietary or commercial data sources. Furthermore, satellite observations originate from sensors with differing spectral characteristics, resolutions, and retrieval methodologies. The integration of multispectral and hyperspectral data, as well as data from heterogeneous sensor technologies, introduces additional complexity and potential inconsistencies in the feature space. Finally, the absence of comprehensive ground-truth emission inventories limits the ability to perform fully quantitative validation. As a result, model evaluation is primarily based on proxy labels and relative performance metrics rather than absolute emission estimates.

VI. Conclusion and Future Work

This study presents a scalable and robust framework for methane hotspot detection based on satellite observations and machine learning. By integrating multi-source data within a consistent spatiotemporal representation and by combining leakage-free preprocessing with spatially aware validation, the proposed approach enables reliable detection across diverse geographic regions. The results demonstrate that methane hotspot detection is primarily driven by direct atmospheric observations, with methane-derived features acting as the dominant predictors. Environmental variables provide complementary contextual information but are insufficient for accurate prediction in isolation. These findings highlight both the strengths and limitations of data-driven approaches relying on indirect proxies. Overall, the proposed framework establishes a foundation for large-scale, satellite-based methane monitoring. It further provides a structured basis for future advancements, including the integration of higher-resolution and hyperspectral datasets, as well as improved modeling of environmental and atmospheric influences on methane dynamics. Future work will incorporate additional environmental variables, improve temporal resolution and integrate higher-resolution datasets such as EMIT. In particular, hyperspectral data offers the potential for more precise plume detection and source attribution. Additionally, advanced clustering and spatial analysis methods will be explored to improve the identification and characterization of emission zones.

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